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Inventor(s)	Yakes; Christopher K. et al.

Refuse vehicle with independently operational accessory system

Abstract

A refuse vehicle includes multiple tractive elements, a prime mover, and an independent accessory system. The prime mover is configured to generate mechanical energy to drive one or more of the tractive elements. The independent accessory system includes one or more storage tanks configured to store a fuel, and an accessory primary mover. The accessory primary mover is configured to fluidly couple with the one or more storage tanks to receive the fuel from the one or more storage tanks and operate to pressurize a hydraulic fluid to drive an accessory of the refuse vehicle. The accessory primary mover is configured to pressurize the hydraulic fluid to drive the accessory of the refuse vehicle independently of operation of the prime mover.

Inventors: Yakes; Christopher K. (Oshkosh, WI), Nasr; Nader (Neenah, WI), Messina; Robert S. (Oshkosh, WI), Rocholl; Joshua D. (Rochester, MN), Wentz; Derek A. (Austin, MN), Hoover; Vincent (Byron, MN), Klein; Zachary L. (Rochester, MN), Weckwerth; Clinton T. (Pine Island, MN), Wachter; Skylar A. (Dodge Center, MN), Koga; Jeff (Oshkosh, WI), Davis; Emily (Rochester, MN)

Applicant: Oshkosh Corporation (Oshkosh, WI)

Family ID: 74258545

Assignee: Oshkosh Corporation (Oshkosh, WI)

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Primary Examiner: Shaawat; Mussa A

Attorney, Agent or Firm: Foley & Lardner LLP

Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 18/131,701, filed Apr. 6, 2023, which is a continuation of U.S. application Ser. No. 17/221,255, filed Apr. 2, 2021, which is a continuation of U.S. application Ser. No. 17/007,605, filed Aug. 31, 2020, now U.S. Pat. No. 11,001,135 which is a continuation of U.S. application Ser. No. 16/943,295, filed Jul. 30, 2020, now U.S. Pat. No. 11,007,863 which

BACKGROUND

(1) Refuse vehicles collect a wide variety of waste, trash, and other material from residences and businesses. Operators of the refuse vehicles transport the material from various waste receptacles within a municipality to a storage or processing facility (e.g., a landfill, an incineration facility, a recycling facility, etc.).

SUMMARY

(2) One embodiment of the present disclosure relates to a refuse vehicle. The refuse vehicle includes multiple tractive elements, a prime mover, and an independent accessory system. The prime mover is configured to generate mechanical energy to drive one or more of the tractive elements. The independent accessory system includes one or more storage tanks configured to store a fuel, and an accessory primary mover. The accessory primary mover is configured to fluidly couple with the one or more storage tanks to receive the fuel from the one or more storage tanks and operate to pressurize a hydraulic fluid to drive an accessory of the refuse vehicle. The accessory primary mover is configured to pressurize the hydraulic fluid to drive the accessory of the refuse vehicle independently of operation of the prime mover.

(3) Another embodiment of the present disclosure relates to an independent accessory system for a refuse vehicle. The system includes one or more storage tanks configured to store a fuel, and an accessory primary mover. The accessory primary mover is configured to fluidly couple with the one or more storage tanks to receive the fuel from the one or more storage tanks and operate to pressurize a hydraulic fluid to drive an accessory of the refuse vehicle. The accessory primary mover is configured to pressurize the hydraulic fluid to drive the accessory of the refuse vehicle independently of operation of a prime mover of the refuse vehicle.

(4) Another embodiment of the present disclosure relates to a refuse vehicle. The refuse vehicle includes an independent compressed natural gas (CNG) system including multiple CNG storage tanks configured to store CNG fuel. The independent compressed natural gas (CNG) also includes an independent CNG engine. The independent CNG engine is configured to receive the CNG fuel from the multiple CNG storage tanks and generate mechanical energy using the CNG fuel. The independent CNG engine operates independently of operation of a primary mover of the refuse vehicle.

(5) This summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices or processes described herein will become apparent in the detailed description set forth herein, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a refuse vehicle including an independent accessory system, according to an exemplary embodiment.

(2) FIG. 2 is a block diagram of the independent accessory system of the refuse vehicle of FIG. 1, according to an exemplary embodiment.

(3) FIG. 3 is a block diagram of a control system of the refuse vehicle of FIG. 1, according to an exemplary embodiment.

(4) FIG. 4 is a block diagram of a charging system for the refuse vehicle of FIG. 1, according to an exemplary embodiment.

(5) FIG. 5 is a perspective view of a support structure for fuel tanks of the independent accessory system of the refuse vehicle of FIG. 1, according to an exemplary embodiment.

(6) FIG. 6 is a perspective view of the refuse vehicle of FIG. 1, with an accessory power unit that contains some of the components of the independent accessory system of FIG. 1, according to an exemplary embodiment.

DETAILED DESCRIPTION

(7) Before turning to the FIGURES, which illustrate certain exemplary embodiments in detail, it should be understood that the present disclosure is not limited to the details or methodology set forth in the description or illustrated in the FIGURES. It should also be understood that the terminology used herein is for the purpose of description only and should not be regarded as limiting.

Overview

(8) Referring generally to the FIGURES, a refuse vehicle includes a prime mover configured to drive the refuse vehicle for transportation. The refuse vehicle may include tractive elements (e.g., wheels) that are configured to be driven by the prime mover to transport the refuse vehicle from location to location. The prime mover can be an electric motor, a compressed natural gas (CNG) engine, an internal combustion engine (e.g., a diesel engine, a gasoline engine, etc.), or any combination thereof. For example, the refuse vehicle may be a hybrid refuse vehicle that includes both an electric motor and an internal combustion engine.

(9) The refuse vehicle also includes an independent accessory system that is configured to operate various body functions of the refuse vehicle. For example, the independent accessory system can be configured to operate lift arms, a packer apparatus, a tailgate, lifting/dumping apparatuses, etc., of the refuse vehicle. The independent accessory system can include one or more fuel tanks (e.g., pressure vessels) that store fuel (e.g., CNG fuel, diesel fuel, gasoline fuel, etc.) for use by an engine (e.g., an internal combustion engine). The fuel may be stored in the one or more fuel tanks as a liquid fuel, a gaseous fuel, or a combination thereof (e.g., a saturated fuel). The engine may be configured to fluidly couple with the fuel tanks to receive fuel from the tanks, combust the fuel, and drive a hydraulic pump. The hydraulic pump can draw or recirculate hydraulic fluid from a reservoir and provide the hydraulic fluid to one or more hydraulic cylinders. The hydraulic cylinders can be operated to perform various body functions (e.g., by extending and/or retracting).

(10) The independent accessory system can be operated by a user through a human machine interface (HMI) and a controller. The controller may receive user inputs from the HMI and generate control signals for the engine and/or the hydraulic pump to perform requested operations of the body functions. The engine and the hydraulic pump may be sized according to requirements of the various body functions. For example, a compaction apparatus that compacts, crushes, compresses, or otherwise packs refuse may require a larger hydraulic cylinder, hydraulic pump, and engine. Likewise, a smaller hydraulic cylinder, hydraulic pump, and engine may be suitable for lift arms for small refuse collection bins.

(11) In some embodiments, one or more of the components of the independent accessory system are positioned within a modular unit (e.g., a modular add-on unit, an accessory power unit, etc.). The modular unit can be removably coupled with the refuse vehicle. The modular unit can include the engine, the hydraulic pump, a reservoir for the hydraulic pump, etc. In some embodiments, the modular unit is configured to fluidly couple with the fuel tanks to receive the fuel from the fuel tanks. The modular unit can be removably and/or fixedly coupled anywhere on the refuse vehicle, and may be fluidly coupled with the fuel tanks.

(12) The prime mover of the refuse vehicle may be an electric motor. If the prime mover is an electric motor, the refuse vehicle may include a battery system having battery cells. The battery cells may store electrical energy (e.g., in the form of chemical energy) and provide the electrical energy to the electric motor for transportation. The battery system can be configured to removably electrically couple with a charging station that may be located at jobsites, along a route of the refuse vehicle, at charging locations, at a fleet management location (e.g., a home base), etc. The charging station can include an engine, a generator, and fuel tanks. The fuel tanks can provide the

engine with fuel. The engine combusts the fuel and drives the generator (e.g., through a shaft). The generator then charges the batteries with electrical power/electrical energy that can be used to transport the refuse vehicle.

Overall Vehicle

(13) As shown in FIG. 1, a vehicle, shown as refuse vehicle **10** (e.g., a garbage truck, a waste collection truck, a sanitation truck, a recycling truck, etc.), is configured as a front-loading refuse truck. In other embodiments, the refuse vehicle **10** is configured as a side-loading refuse truck or a rear-loading refuse truck. In still other embodiments, the vehicle is another type of vehicle (e.g., a skid-loader, a telehandler, a plow truck, a boom lift, etc.). As shown in FIG. 1, the refuse vehicle **10** includes a chassis, shown as frame **12**; a body assembly, shown as body **14**, coupled to the frame **12** (e.g., at a rear end thereof, etc.); and a cab, shown as cab **16**, coupled to the frame **12** (e.g., at a front end thereof, etc.). The cab **16** may include various components to facilitate operation of the refuse vehicle **10** by an operator (e.g., a seat, a steering wheel, actuator controls, a user interface, switches, buttons, dials, etc.).

(14) As shown in FIG. 1, the refuse vehicle **10** includes an electric motor, a CNG engine, a hybrid engine, an internal combustion engine, a diesel engine, a gasoline engine, etc., shown as prime mover **18**, and an energy storage system, shown as battery system **20**. In some embodiments, the prime mover is or includes an internal combustion engine. For example, the prime mover may be a diesel engine, a gasoline engine, a CNG engine, etc. According to the exemplary embodiment shown in FIG. 1, the prime mover **18** is coupled to the frame **12** at a position beneath the cab **16**. The prime mover **18** is configured to provide power to a plurality of tractive elements, shown as wheels **22** (e.g., via a drive shaft, axles, etc.). In other embodiments, the prime mover **18** is otherwise positioned and/or the refuse vehicle **10** includes a plurality of electric motors to facilitate independently driving one or more of the wheels **22**. In still other embodiments, the prime mover **18** or a secondary electric motor is coupled to and configured to drive a hydraulic system that powers hydraulic actuators. According to the exemplary embodiment shown in FIG. 1, the battery system **20** is coupled to the frame **12** beneath the body **14**. In other embodiments, the battery system **20** is otherwise positioned (e.g., within a tailgate of the refuse vehicle **10**, beneath the cab **16**, along the top of the body **14**, within the body **14**, etc.).

(15) According to an exemplary embodiment, the battery system **20** is configured to provide electric power to (i) the prime mover **18** to drive the wheels **22**, (ii) electric actuators of the refuse vehicle **10** to facilitate operation thereof (e.g., lift actuators, tailgate actuators, packer actuators, grabber actuators, etc.), and/or (iii) other electrically operated accessories of the refuse vehicle **10** (e.g., displays, lights, etc.). In some embodiments, the refuse vehicle **10** includes an internal combustion generator that utilizes one or more fuels (e.g., gasoline, diesel, propane, natural gas, hydrogen, etc.) to generate electricity to charge the battery system **20**, power the prime mover **18**, power the electric actuators, and/or power the other electrically operated accessories (e.g., a hybrid refuse vehicle, etc.). For example, the refuse vehicle **10** may have an internal combustion engine augmented by the prime mover **18** to cooperatively provide power to the wheels **22**. The battery system **20** may thereby be charged via an on-board generator (e.g., an internal combustion generator, a solar panel system, etc.), from an external power source (e.g., overhead power lines, mains power source through a charging input, etc.), and/or via a power regenerative braking system, and provide power to the electrically operated systems of the refuse vehicle **10**.

(16) According to an exemplary embodiment, the refuse vehicle **10** is configured to transport refuse from various waste receptacles within a municipality to a storage and/or processing facility (e.g., a landfill, an incineration facility, a recycling facility, etc.). As shown in FIG. 1, the body **14** includes a plurality of panels, shown as panels **32**, a tailgate **34**, and a cover **36**. The panels **32**, the tailgate **34**, and the cover **36** define a collection chamber (e.g., hopper, etc.), shown as refuse compartment **30**. Loose refuse may be placed into the refuse compartment **30** where it may thereafter be compacted (e.g., by a packer system, etc.). The refuse compartment **30** may provide temporary

storage for refuse during transport to a waste disposal site and/or a recycling facility. In some embodiments, at least a portion of the body **14** and the refuse compartment **30** extend above or in front of the cab **16**. According to the embodiment shown in FIG. **1**, the body **14** and the refuse compartment **30** are positioned behind the cab **16**. In some embodiments, the refuse compartment **30** includes a hopper volume and a storage volume. Refuse may be initially loaded into the hopper volume and thereafter compacted into the storage volume by a compacting apparatus **46**. According to an exemplary embodiment, the hopper volume is positioned between the storage volume and the cab **16** (e.g., refuse is loaded into a position of the refuse compartment **30** behind the cab **16** and stored in a position further toward the rear of the refuse compartment **30**, a front-loading refuse vehicle, a side-loading refuse vehicle, etc.). In other embodiments, the storage volume is positioned between the hopper volume and the cab **16** (e.g., a rear-loading refuse vehicle, etc.).

(17) As shown in FIG. **1**, the refuse vehicle **10** includes a lift mechanism/system (e.g., a front-loading lift assembly, etc.), shown as lift assembly **40**, coupled to the front end of the body **14**. In other embodiments, the lift assembly **40** extends rearward of the body **14** (e.g., a rear-loading refuse vehicle, etc.). In still other embodiments, the lift assembly **40** extends from a side of the body **14** (e.g., a side-loading refuse vehicle, etc.). As shown in FIG. **1**, the lift assembly **40** is configured to engage a container (e.g., a residential trash receptacle, a commercial trash receptacle, a container having a robotic grabber arm, etc.), shown as refuse container **60**. The lift assembly **40** may include various actuators (e.g., electric actuators, hydraulic actuators, pneumatic actuators, etc.), shown as hydraulic cylinders **108**, to facilitate engaging the refuse container **60**, lifting the refuse container **60**, and tipping refuse out of the refuse container **60** into the hopper volume of the refuse compartment **30** through an opening in the cover **36** or through the tailgate **34**. The lift assembly **40** may thereafter return the empty refuse container **60** to the ground. According to an exemplary embodiment, a door, shown as top door **38**, is movably coupled along the cover **36** to seal the opening thereby preventing refuse from escaping the refuse compartment **30** (e.g., due to wind, bumps in the road, etc.).

Accessory Power System

(18) Referring still to FIG. **1**, the refuse vehicle **10** also includes an independent accessory system **100** (e.g., a CNG powered accessory system, a diesel powered accessory system, etc.), according to an exemplary embodiment. The independent accessory system **100** can be configured to drive, move, provide mechanical energy for, etc., or otherwise operate various body functions of refuse vehicle **10** independently of an operation of prime mover **18**. For example, the independent accessory system **100** can be configured to drive or operate the lift assembly **40**, a tailgate lift assembly **42**, etc., or any other body function, lift apparatus, auxiliary apparatus, etc., of the refuse vehicle **10**. In some embodiments, the independent accessory system **100** is configured to operate a hydraulic cylinder **108** of any of the lift apparatuses, auxiliary apparatuses, etc. The independent accessory system **100** may be configured to operate independently of the prime mover (e.g., prime mover **18**) of the refuse vehicle **10**. In some embodiments, the independent accessory system **100** can operate to drive the hydraulic cylinders **108** without requiring operation of the prime mover **18**. For example, the independent accessory system **100** can independently provide mechanical energy for the various body functions **114** of the refuse vehicle **10**, without requiring operation of or mechanical energy from the prime mover **18** (e.g., even if prime mover **18** is shut-off or inoperational, or in an idle mode). In some embodiments, the independent accessory system **100** and operation of the prime mover **18** are linked (e.g., linked in a control scheme). However, if operation of the independent accessory system **100** and the prime mover **18** are linked in a control scheme, the independent accessory system **100** and the prime mover **18** (e.g., the prime mover of the refuse vehicle **10**) may still be able to provide mechanical energy for their respective functions (e.g., operation of the body functions **114** and transportation of the refuse vehicle **10**, respectively) independent of the operation of each other.

(19) The body functions can include operation of lift arms (e.g., front loading lift arms, side loading

lift arms, rear loading lift arms), tailgates, dumping operations, packing operations, etc., of the refuse vehicle **10**. The refuse vehicle **10** can include various hydraulic cylinders **108** configured to perform any of the body functions described herein. For example, the refuse vehicle **10** can include the compacting apparatus **46** that is configured to pack, crush, compact, compress, etc., refuse that is loaded into the hopper or the body **14** using the hydraulic cylinders **108**. The independent accessory system **100** can be configured to operate any of the hydraulic cylinders **108** to perform the various body functions in response to user inputs. The independent accessory system **100** can be configured to perform the various body functions independently of each other, or in conjunction with each other.

(20) Referring to FIGS. **1** and **2**, the independent accessory system **100** includes one or more tanks, capsules, containers, pressure vessels, cartridges, etc., shown as fuel tanks **104** (e.g., CNG tanks, diesel fuel tanks, gasoline tanks, etc.). The fuel tanks **104** are supported, fixedly coupled, fixed, connected, etc., or otherwise coupled with a support unit, a mount unit, a structure, etc., shown as support structure **102** of the refuse vehicle **10**. In some embodiments, the fuel tanks **104** are positioned within the tailgate **34** (e.g., as shown in FIG. **5**, described in greater detail below). For example, the fuel tanks **104** and the support structure **102** can be disposed within an inner volume of the tailgate **34**.

(21) The independent accessory system **100** also includes an internal combustion engine, a CNG engine, a diesel engine, a fuel cell, a hydrogen engine, an electric motor, etc., shown as accessory prime mover **110**. The accessory prime mover **110** is configured to receive fuel (e.g., diesel fuel, gasoline, CNG, hydrogen, electrical energy, a resource, etc.) from the fuel tanks **104** through a piping system, a plumbing system, one or more pipes, etc. The piping system can include various tubular members, pipes, hoses, valves, connectors, etc., that fluidly couple with the tank **112** and the accessory prime mover **110** such that fuel can be provided from the tank **112** to the accessory prime mover **110**. The accessory prime mover **110** can use the fuel (e.g., combust the fuel) to produce mechanical energy. The mechanical energy is output by the accessory prime mover **110** to a pump **106**. The pump **106** can be driven by the accessory prime mover **110** and draw hydraulic fluid from a fluid reservoir, a tank, etc., shown as tank **112**. The tank **112** is coupled with (e.g., fixedly coupled, attached, mounted, etc.) with the refuse vehicle **10**. The tank **112** can be fixedly coupled with the body **14**. The pump **106** outputs the hydraulic fluid to the hydraulic cylinders **108** to operate the hydraulic cylinders **108** to perform the body functions **114**.

(22) In some embodiments, the accessory prime mover **110** is a smaller engine than the prime mover **18**. The accessory prime mover **110** and the hydraulic pump **106** can be sized according to requirements of the various body functions. Other refuse vehicles use the prime mover **18** to drive the body functions. However, this may be inefficient, since the prime mover **18** is sized to transport the refuse vehicle **10** (e.g., to provide torque to the wheels **22**) and may be oversized for the body functions. Using a smaller engine (e.g., the accessory prime mover **110**) with a correspondingly sized hydraulic pump **106** facilitates a more efficient and robust refuse vehicle, which does not use an oversized prime mover **18** for body functions.

(23) Advantageously, the independent accessory system **100** can use pre-existing infrastructure of the refuse vehicle **10**. For example, CNG-powered refuse vehicles (e.g., refuse vehicles that use a CNG engine as the prime mover for transportation purposes) may already include a support structure and fuel tanks that can be used by the accessory prime mover **110**/hydraulic pump **106** for the body functions.

Accessory Power Unit

(24) Referring particularly to FIG. **6**, one or more portions of the independent accessory system **100** can be contained in, enclosed in, supported by, etc., a modular unit, an add-on unit, a removable unit, etc., shown as accessory power unit (APU) **120**. The APU **120** can be configured to integrate with existing infrastructure (e.g., CNG infrastructure) of the refuse vehicle **10** to operate or drive the various body functions of the refuse vehicle **10**. The refuse vehicle **10** can be

configured as a front loading refuse vehicle, a side loading refuse vehicle, a rear loading refuse vehicle, etc. It should be understood that while the inventive concepts described herein reference a refuse vehicle, it is contemplated that the APU **120** and/or the various components of the independent accessory system **100** are also applicable to various other types of vehicles that include body functions. For example, the independent accessory system **100** and/or the APU **120** can be used on a fire truck, a commercial truck, a heavy-duty truck, etc., or any vehicle that has body functions to be operated independently of the transportation of the vehicle.

(25) The APU **120** can be removably coupled with the refuse vehicle **10** on an underside of the body **14**. For example, the APU **120** can be fixedly and removably coupled with the frame **12** beneath the body **14**. The APU **120** can be fixedly and removably coupled at a front of the frame **12**, at a rear end of the frame **12**, centrally along the frame **12**, etc. In other embodiments, the APU **120** can be fixedly and removably coupled with a side of the body **14**, within the body **14**, within a compartment of the body **14**, on top of the body **14**, etc. The APU **120** can be positioned anywhere about the body **14** or anywhere on the refuse vehicle **10** that provides sufficient structural strength (e.g., along the frame **12**, near a chassis of the refuse vehicle **10**, etc.).

(26) The APU **120** includes the accessory prime mover **110**, the tank **112**, and the hydraulic pump **106**, according to an exemplary embodiment. The APU **120** can be a hollow container that protects the various internal components (e.g., the accessory prime mover **110**, the tank **112**, the hydraulic pump **106**, etc.) and removably couples with the refuse vehicle **10**. The accessory prime mover **110** of the APU **120** fluidly couples with the fuel tanks **104** through a plumbing system, a piping system, etc., shown as tubular system **122**. The tubular system **122** includes various tubular members that fluidly couple the fuel tanks **104** with the accessory prime mover **110**. The accessory prime mover **110** receives the fuel from the fuel tanks **104** through the tubular system **122**, combusts the fuel, and drives the hydraulic pump **106**. The hydraulic pump **106** then drives the hydraulic cylinder(s) **108** of the various body functions of the refuse vehicle **10** (e.g., through various tubular members, pipes, etc.).

(27) Advantageously, the APU **120** facilitates a versatile refuse vehicle with improved efficiency since the accessory prime mover **110** and the hydraulic pump **106** are sized to serve or drive the various hydraulic cylinders **108**. The APU **120** can be installed by a technician, plumbed (e.g., by fluidly coupling the accessory prime mover **110** with the fuel tanks **104** through installation of the tubular system **122**), and used to operate the various body functions of the refuse vehicle **10**. Advantageously, the various body functions of the refuse vehicle **10** can be operated independently of the prime mover **18**. The APU **120** can integrate with existing structure (e.g., existing fuel tanks **104**), to thereby convert refuse vehicles to the refuse vehicle **10** described herein.

Control System

(28) Referring particularly to FIG. 3, a control system **200** can be configured to operate the refuse vehicle **10**, according to an exemplary embodiment. The control system **200** includes a controller that is configured to generate control signals for a drivetrain, a chassis, etc., of the refuse vehicle **10**, shown as drivetrain **210**. The drivetrain **210** includes an engine **202**, a transmission **204**, and wheels **22** of the refuse vehicle **10**. The engine **202** may be the prime mover **18** of the refuse vehicle **10**. The engine **202** can produce mechanical energy and output the mechanical energy to the transmission **204**. The transmission **204** receives the mechanical energy from the engine **202** and outputs mechanical energy (e.g., rotational kinetic energy) to the wheels **22** (e.g., at a higher torque than the mechanical energy input by the engine **202**).

(29) Control system **200** includes a controller **208** that is configured to generate control signals for the engine **202** and the transmission **204**. The controller **208** can include a circuit, shown as processing circuit **216**, a processor, shown as processor **212**, and memory, shown as memory **214**, according to an exemplary embodiment. Controller **208** may be implemented as a general-purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a digital-signal-processor (DSP), circuits containing one or more processing

components, circuitry for supporting a microprocessor, a group of processing components, or other suitable electronic processing components. The processing circuit **216** of controller **208** may include an ASIC, one or more FPGAs, a DSP, circuits containing one or more processing components, circuitry for supporting a microprocessor, a group of processing components, or other suitable electronic processing components (i.e., processor **212**). In some embodiments, the processing circuit **216** is configured to execute computer code stored in memory **214** to facilitate the activities described herein.

(30) Memory **214** may be any volatile or non-volatile computer-readable storage medium capable of storing data or computer code relating to the activities described herein. According to an exemplary embodiment, memory **214** includes computer code modules (e.g., executable code, object code, source code, script code, machine code, etc.) configured for execution by the processing circuit **216**.

(31) In some embodiments, a single controller **208** is configured to generate control signals for both the drivetrain **210** and the independent accessory system **100**. In other embodiments, multiple controllers **208** are configured to generate control signals for independent accessory system **100** and drivetrain **210** independently of each other. For example, a first controller **208** can be configured to provide control signals to engine **202** and/or transmission **204** of drivetrain **210**, while a second controller **208** can be configured to provide control signals for independent accessory system **100**. The first and second controllers **208** can be configured to receive user inputs from a human machine interface (HMI) or a user interface, shown as HMI **218**. In some embodiments, the first and second controllers **208** are configured to receive user inputs from separate HMIs **218**. The HMIs **218** can be positioned within the cab **16** or near the associated body functions (e.g., near lift assembly **40**). For example, the HMI **218** that controls the operation of the drivetrain **210** can be disposed within the cab **16**, while the HMI **218** that controls the operation of the lift assembly **40** can be positioned on the body **14** near the lift assembly **40**.

Charging System

(32) Referring particularly to FIG. **4**, a charging system **300** can be used to re-charge batteries **44** of the battery system **20**, according to an exemplary embodiment. The charging system **300** includes a charging station **302** that can be positioned at a fleet management site, at a job site, along the refuse vehicle's route, etc. The charging station **302** includes one or more fuel tanks **306**, an engine **304**, and a generator **308**. The fuel tanks **306** can be the same as or similar to the fuel tanks **104** on the refuse vehicle **10**. Likewise, the engine **304** can be similar to the accessory prime mover **110** on the refuse vehicle **10**.

(33) The refuse vehicle **10** includes controller **208** that is configured to generate and provide control signals for prime mover **18** (e.g., an electric motor) and/or accessory prime mover **110**. The controller **208** can be configured to receive user inputs from HMI **218** and generate the control signals for the prime mover **18** and/or the accessory prime mover **110** based on the user inputs. In some embodiments, the controller **208** generates control signals to operate the prime mover **18** and/or the accessory prime mover **110** to perform operations requested by the user through HMI **218**.

(34) The refuse vehicle **10** can be driven by an electric motor, an engine (e.g., engine **202**), or a hybrid engine-electric motor. In this way, the refuse vehicle may be an electrically driven refuse vehicle, an internal-combustion engine driven vehicle, or a hybrid vehicle. For example, the refuse vehicle **10** can include a plurality of prime movers. One or more of the prime movers can be electric motors (e.g., the prime mover **18**) and/or internal combustion engines (e.g., the engine **202**). The electric motors used to transport the refuse vehicle **10** are supplied with power by batteries **44** of the battery system **20**.

(35) The batteries **44** can be removable and/or replaceable battery cells. For example, the batteries **44** can be charged at a fleet management site in a charging rack, then installed into the refuse vehicle **10**. The batteries **44** can be later removed (e.g., after a state of charge of the batteries **44** has

been depleted) and replaced with new or fresh batteries (e.g., that may be stored on the refuse vehicle **10**).

(36) The operator of the refuse vehicle **10** may arrive at a job site, or at a fleet management location and electrically couple the charging station **302** with the batteries **44**. Since some refuse vehicles operate using CNG, the charging station **302** may use pre-existing fuel tanks **306** at the fleet management location that store CNG. In some embodiments, the engine **304** and the generator **308** are packaged in a unit that is configured to fluidly couple with the fuel tanks **306**. In some embodiments, the fuel tanks **306** are removably fluidly coupled with the engine **304**. In this way, the fuel tanks **306** can be used for replenishing the fuel tanks **104** on the refuse vehicle **10** and/or for charging batteries **44** of the refuse vehicle **10**. The generator **308** can be any mechanical transducer capable of receiving mechanical energy (e.g., rotational kinetic energy) and generating electrical energy for the batteries **44**. For example, the generator **308** can include a stator and an armature that is driven by the engine **304** to produce electrical current or electrical energy.

(37) Referring again to FIG. **2**, the accessory prime mover **110** can be configured to output mechanical energy to generator **116** to drive generator **116** to generate electrical power. The electrical power is provided to batteries **44** of refuse vehicle **10** to charge the batteries **44**. In this way, accessory prime mover **110** can operate independently to drive generator **116** to charge batteries **44** of refuse vehicle **10**.

Support Infrastructure

(38) Referring now to FIG. **5**, one possible infrastructure of the support structure **102** includes the fuel tanks **104** stored within the tailgate **34**. The tailgate **34** can include a first or inner member **502** and a second or outer member **504**. The first member **502** is configured to fixedly and/or pivotally couple with the refuse vehicle **10**. The first member **502** and the second member **504** can be configured to removably and fixedly couple with each other to define an inner volume. The fuel tanks **104** can be fixedly coupled with the first member **502** and stored within the inner volume defined by the first member **502** and the second member **504**. The fuel tanks **104** can be oriented horizontally (as shown in FIG. **5**) or vertically. The fuel tanks **104** can be fixedly coupled with the first member **502** at their ends (e.g., with fasteners). In some embodiments, the fuel tanks **104** extend along substantially an entire width of the tailgate **34**.

(39) It should be understood that while several configurations of the support structure **102** are described herein, the inventive concepts are not limited to these configurations of the support structure **102**. The fuel tanks **104** can be positioned anywhere on the refuse vehicle **10**, or in multiple locations. For example, the fuel tanks **104** can be positioned on top of the refuse vehicle **10** (e.g., on top of the body **14**), underneath the refuse vehicle **10** (e.g., on an underside of the body **14**, on the frame **12** beneath the refuse vehicle **10**, etc.), between the cab **16** and the body **14**, etc.

Configuration of Exemplary Embodiments

(40) As utilized herein, the terms “approximately,” “about,” “substantially”, and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

(41) It should be noted that the term “exemplary” and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, or illustrations of possible embodiments (and such terms are not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

(42) The term “coupled” and variations thereof, as used herein, means the joining of two members

directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

(43) References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below”) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

(44) The hardware and data processing components used to implement the various processes, operations, illustrative logics, logical blocks, modules and circuits described in connection with the embodiments disclosed herein may be implemented or performed with a general purpose single-or multi-chip processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor may be a microprocessor, or, any conventional processor, controller, microcontroller, or state machine. A processor also may be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. In some embodiments, particular processes and methods may be performed by circuitry that is specific to a given function. The memory (e.g., memory, memory unit, storage device) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present disclosure. The memory may be or include volatile memory or non-volatile memory, and may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. According to an exemplary embodiment, the memory is communicably connected to the processor via a processing circuit and includes computer code for executing (e.g., by the processing circuit or the processor) the one or more processes described herein.

(45) The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for

example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

(46) Although the figures and description may illustrate a specific order of method steps, the order of such steps may differ from what is depicted and described, unless specified differently above. Also, two or more steps may be performed concurrently or with partial concurrence, unless specified differently above. Such variation may depend, for example, on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations of the described methods could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

(47) It is important to note that the construction and arrangement of the refuse vehicle **10** and the systems and components thereof as shown in the various exemplary embodiments is illustrative only. Additionally, any element disclosed in one embodiment may be incorporated or utilized with any other embodiment disclosed herein. Although only one example of an element from one embodiment that can be incorporated or utilized in another embodiment has been described above, it should be appreciated that other elements of the various embodiments may be incorporated or utilized with any of the other embodiments disclosed herein.

Claims

1. A refuse vehicle, comprising: a prime mover configured to generate mechanical energy; and an independent accessory system, comprising: an accessory primary mover configured to receive and operate to pressurize a hydraulic fluid to drive an accessory of the refuse vehicle; wherein the accessory primary mover is configured to pressurize the hydraulic fluid to drive the accessory of the refuse vehicle independently of operation of the prime mover.
2. The refuse vehicle of claim 1, wherein the prime mover is at least one of an internal combustion engine, a compressed natural gas engine, or an electric motor.
3. The refuse vehicle of claim 2, further comprising: a battery system comprising a plurality of battery cells configured to store energy, wherein the electric motor is configured to draw electrical power from the plurality of battery cells to drive one or more of a plurality of tractive elements.
4. The refuse vehicle of claim 3, wherein the accessory primary mover is configured to drive a generator to charge the plurality of battery cells.
5. The refuse vehicle of claim 4, wherein the plurality of battery cells are driven to charge by the generator, the generator being driven by another accessory primary mover separate from the accessory primary mover used to pressurize the hydraulic fluid.
6. The refuse vehicle of claim 5, wherein the generator, the other accessory primary mover, and a plurality of charging fuel tanks are stored at a charging station, wherein the refuse vehicle is configured to removably electrically couple with the generator of the charging station to charge the plurality of battery cells of the refuse vehicle.
7. The refuse vehicle of claim 1, wherein the independent accessory system is a compressed natural gas (CNG) system, wherein a fuel of the independent accessory system is a compressed natural gas, and the accessory primary mover is a CNG engine configured to consume the compressed natural gas to drive a hydraulic pump.
8. The refuse vehicle of claim 1, wherein the independent accessory system is a diesel system, wherein a fuel of the independent accessory system is a diesel fuel, and the accessory primary mover is an internal combustion engine configured to consume the diesel fuel to drive a hydraulic pump.
9. The refuse vehicle of claim 1, wherein the accessory primary mover and a hydraulic pump that is driven by the accessory primary mover to pressurize the hydraulic fluid are positioned in a modular unit, wherein the modular unit is removably coupled with the refuse vehicle.

10. The refuse vehicle of claim 9, wherein the modular unit is a tailgate that is coupled with the refuse vehicle.

11. An independent accessory system for a vehicle, the independent accessory system comprising: an accessory primary mover configured to receive fuel and operate to drive an accessory of the vehicle; wherein the accessory primary mover is configured to drive the accessory of the vehicle independently of operation of a prime mover of the vehicle.

12. The independent accessory system of claim 11, wherein the prime mover of the vehicle is at least one of an internal combustion engine, a compressed natural gas engine, or an electric motor.

13. The independent accessory system of claim 11, wherein the accessory primary mover is configured to drive a generator to generate electrical energy to charge battery cells of the vehicle.

14. The independent accessory system of claim 11, wherein the independent accessory system is a compressed natural gas (CNG) system, wherein the fuel is a compressed natural gas, and the accessory primary mover is a CNG engine configured to consume the compressed natural gas to drive a hydraulic pump.

15. The independent accessory system of claim 11, wherein the independent accessory system is a diesel system, wherein the fuel is a diesel fuel and the accessory primary mover is an internal combustion engine configured to consume the diesel fuel to drive a hydraulic pump.

16. The independent accessory system of claim 11, wherein the accessory primary mover and a hydraulic pump that is driven by the accessory primary mover to pressurize a hydraulic fluid are positioned in a modular unit, wherein the modular unit is removably coupled with the vehicle.

17. The independent accessory system of claim 16, wherein the modular unit is a tailgate of the vehicle.

18. A vehicle, comprising: an independent compressed natural gas (CNG) system comprising: a CNG storage tank configured to store CNG fuel; and an independent CNG engine configured to receive the CNG fuel from the CNG storage tank and generate mechanical energy using the CNG fuel, wherein the independent CNG engine operates independently of operation of a primary mover of the vehicle.

19. The vehicle of claim 18, wherein the independent CNG engine is configured to drive operation of one or more accessories of the vehicle.

20. The vehicle of claim 18, wherein the independent CNG engine is configured to drive a generator to charge a battery of the vehicle.
